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Research Article

A multiprofessional intervention for patients at risk of physical and psychological problems after ICU stay: A feasibility study

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ABSTRACT

Objectives: This study aimed to evaluate the feasibility and fidelity of a multiprofessional intervention for intensive care survivors at increased risk of physical and/or psychological problems.
Methods: The study was conducted in two ICUs, with a single-arm interventional design. Adult patients (≥18 years) with an ICU length of stay of>12 hours were assessed using a combined validated screening instrument to estimate the risk of physical and/or psychological morbidity. The intervention consisted of 1) in-hospital follow-up comprising several components, a first ward visit by a case manager within 1–5 days after ICU discharge and establishment and follow-up of an individual rehabilitation plan; 2) out-of-hospital telephone follow-up for 12 weeks. Data on process feasibility (recruitment, retention, and assessment completion) and intervention fidelity (adherence and experiences of intervention delivery) were collected. Intervention acceptability was assessed through interviews with participants and family members, while the in-hospital follow-up programme was evaluated using a questionnaire completed by healthcare professionals and patient-reported outcomes three months after ICU discharge.
Results: Of 155 patients screened for increased risk of long-term sequelae at ICU discharge, 57 were identified as high risk, and 41 consented to participate in the study. The intervention was feasible (retention rate 90%), and acceptable from the perspectives of the participants, family members and healthcare professionals. Fidelity of the intervention components varied, with high adherence to the initial ward visit, development of a rehabilitation plan and involvement of multiple healthcare professionals based on individual needs.
Conclusions: A multiprofessional follow-up programme for high-risk ICU patients is feasible. Feedback informed logistical adjustments while preserving core principles. The findings support a future effectiveness trial, with refinements to improve screening and assessment completion.
Implications for clinical practice: Early multiprofessional follow-up after intensive care appears to be feasible and appreciated by patients at risk of physical and/or psychological problems after ICU stay.

Introduction

More than half of ICU survivors experience long-term physical, psychological and cognitive impairments, collectively referred to as post-intensive care syndrome (PICS) [1,2]. These sequelae significantly hinder patients' ability to return to work and resume daily activities following hospital discharge [3,4]. Moreover, the impact of PICS extends beyond the individual, imposing psychological and emotional burdens on family members [4], representing a growing public health challenge [5]. Transitions from ICU to post-discharge care are frequently perceived by ICU survivors and their families as fragmented, with notable deficiencies in continuity and difficulties navigating their own

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recovery across the healthcare continuum [6]. The persistent symptoms contribute to an ongoing active search for medical attention [7,8]. Among primary-care physicians, awareness of PICS and its implications remains limited [6,9], potentially contributing to fragmented follow-up and unmet patient needs. Several interventions aimed at mitigating PICS have been developed, including ICU follow-up clinics [10,11] and the use of ICU diaries [12]. Other studies have investigated enhanced physical and early rehabilitation programmes targeting physical recovery [13,14]. Although some studies have demonstrated improved patient satisfaction [11,14,15] and certain interventions have been implemented in clinical practice, the evidence remains insufficient to establish their effectiveness in reducing long-term PICS- related impairments. The lack of effect may partly be explained by the inclusion of heterogeneous patient populations, regardless of their individual risk of developing PICS, and variability in the timing of follow-up and delivery of interventions. As a result, it remains unclear which patients are most likely to benefit from follow-up interventions and when such interventions should optimally be initiated. Importantly, most previous interventions have not incorporated systematic risk stratification to target patients at the highest risk of physical and/or psychological sequelae, nor have they been initiated early following ICU discharge. These gaps may contribute to the limited and inconsistent effects observed in prior studies. To address these limitations, we designed a structured multiprofessional follow-up intervention delivered within 1–5 days after ICU discharge and targeted to patients identified as being at increased risk of adverse outcomes. While the intervention was developed with the potential to reduce PICS-related symptoms, this study primarily aimed to evaluate the feasibility of the study procedures, as well as the feasibility and fidelity of the intervention. The findings will inform the design of a subsequent randomised controlled trial.

Methods

Study design

This feasibility study used a single-arm interventional design, appropriate for exploring a new intervention. Given the complexity of evaluating interventions within an individualised context, we employed a design combining quantitative and qualitative approaches. We focused on obtaining participant feedback and evaluating the feasibility of the intervention and study procedures. The study aimed to inform future refinement and scaling of the intervention. The study adhered to the CONSORT 2010 guidelines [16] and Guidelines for reporting non-randomised pilot and feasibility studies [17].

Participants and settings

This study was conducted at two hospitals in Sweden with mixed medical-surgical ICUs from February to May 2024 at one hospital and between October 2024 and March 2025 at the other. Adult patients (≥18 years) with an ICU length of stay of >12 h were considered eligible for inclusion. Exclusion criteria were pre-existing cognitive impairment, structural brain or spinal cord injury, insufficient Swedish proficiency, or no home address. Patients were excluded if they were considered moribund, defined as being in an end-of-life condition with expected imminent death as assessed by the treating clinician. Patients were also excluded if more than one documented treatment limitation was present, defined as ≥2 predefined restrictions of life-sustaining therapy (e.g., limitation of mechanical ventilation, or withholding/withdrawal of organ support) recorded in the medical record.

Recruitment

Eligible patients were consecutively assessed using a combined validated screening instrument to estimate the risk of physical and/or psychological morbidity three months after ICU discharge [18,19]. The screening instrument consists of two parts, one evaluating physical morbidity and the other psychological problems, and is based on established risk factors, including clinical characteristics and aspects of functional status [18,19]. The physical component is based on the first five items of the Chelsea Critical Care Physical Assessment Tool (CPAx) [18,20], while the psychological screening instrument evaluates symptoms of depression and traumatic memories at ICU discharge, perceived social support and age [19]. Each component contributes to an overall estimate of the risk of developing PICS. A predicted probability of ≥20% was defined as indicative of increased risk, and patients exceeding this threshold were included in the study. This cutoff was selected based on the observed non-linear risk curve, where the predicted probability begins to increase more steeply, indicating a transition from low to clinically meaningful risk. This threshold is data-driven and pragmatic, enabling the identification of a subgroup of patients for whom follow-up or preventive interventions may be most relevant. A detailed description of the risk screening instrument and prediction model can be found in Supplementary File 1.

Demographic and clinical data were collected regarding age, sex, Simplified Acute Physiological Score III (SAPS) [21] at ICU admission, mechanical ventilation and ICU length of stay.

Intervention

The intervention protocol was developed using the guidelines for developing complex interventions [22], based on empirical evidence related to PICS [1,2].

The follow-up programme consisted of (Table 1):

In-hospital follow-up.

An ICU nurse or physiotherapist with expertise in PICS and ICU follow-up care acted as case manager and conducted the initial visit on the general ward within 1–5 days following ICU discharge. The follow-up visit included:

- Assessment of delirium using the Confusion Assessment Method for the Intensive Care Unit (CAM-ICU) [23,24] or the Nu-DESC [25,26], both validated instruments for detecting delirium in critically ill patients [27]. Positive screenings led to referral for further cognitive evaluation using the Montreal Cognitive Assessment (MoCA), a commonly used screening tool for cognitive impairment that has been applied in ICU survivors [28,29].
- A structured recapitulation of the ICU course, including treatments and events.
- Handover and review of the ICU diary.
- Assessment of physical function, conducted by a physiotherapist using the 10-item CPAx [(20)] and the Barthel Index [30,31]. The CPAx is a valid and reliable instrument for assessing physical function in critically ill adults [20,32], and has been translated into Swedish [33]. It evaluates ten domains of physical function using a 6-point scale indicating varying levels of independence. The Barthel Index is a reliable and valid instrument for assessing physical functioning at intensive care unit discharge and it assesses ten basic activities of daily living and is a commonly used and reliable measure of physical function at ICU discharge [34], indicating varying levels of independence.
- Self-reported physical activity, using the Swedish National Board of Health and Welfare's two indicator questions (the NBHW-PA questions). One question concerns duration of weekly physical activity and the other addresses duration of weekly physical with response options: 0 min, <30 min, 30–60 min, 60–90 min, 90–120 min, >120 min. Answers are reported as activity minutes (aiming for >150 min/week) or converted into a score where ≥11 corresponds to achieving the World Health Organisation's recommendations [35]. The questionnaire was used to evaluate whether the instrument is feasible for use in a post-intensive care population, as it has not been specifically validated in this population.

Table 1
Intervention components, assessments, and adherence for the 41 participants.

Intervention component	Assessment tools	Time point	Adherence (performed/planned)	Adherence rate (%)
Delirium screening	CAM-ICU or Nu-DESC	1–5 days of ICU discharge	38/41	93%
Cognitive assessment in patients with positive delirium screening	MoCA	Administered within two working days following a positive delirium screening	3/4	75%
Structured recapitulation of ICU stay, including treatments and events	Clinical procedure	1–5 days of ICU discharge	15/41	37%
Handover of ICU diary	Clinical procedure	1–5 days of ICU discharge	13/41	32%
Review of ICU diary	Clinical procedure	1–5 days of ICU discharge	6/41	15%
Assessment of:				
Physical function	CPAx	1–5 days of ICU discharge	9/41	22%
Physical function (in activities of daily living)	Barthel Index	1–5 days of ICU discharge	38/41	93%
Symptoms of anxiety and depression	HADS	1–5 days of ICU discharge	34/41	83%
PTS symptoms	PTSS-14	1–5 days of ICU discharge	31/41	76%
Physical activity	NBHW-PA	1–5 days of ICU discharge	32/41	78%
Pain	NRS	1–5 days of ICU discharge	10/41	24%
Based on the patients' identified needs:				
Rehabilitation plan	Documented in the patient's medical record as goals and sub-goals	1–5 days of ICU discharge	33/41	80%
Referrals to:	Clinical referral	1–5 days of ICU discharge		
Physiotherapy			36/41	
Occupational therapy			5/41	
Counselling support (social worker)			18/41	
Speech therapy			2/41	
Other			4/41	
Telephone follow-up	Structured protocol. Detailed information in Table 2	Once a week for 12 weeks after discharge from ICU	37/40	93%

Abbreviations: CAM-ICU = Confusion Assessment Method for the Intensive Care Unit, Nu-DESC = Nursing Delirium Screening Scale, MoCA = Montreal Cognitive Assessment, ICU = Intensive Care Unit, CPAx = Chelsea Critical Care Physical Assessment Tool, HADS = Hospital Anxiety and Depression Scale, PTS = Post-Traumatic Stress, PTSS-14 = Post-Traumatic Stress Symptoms 14-Question Inventory, NBHW-PA = Swedish National Board of Health and Welfare questions about Physical Activity, NRS = Numeric Rating Scale.

- Assessment of anxiety and depression using the Hospital Anxiety and Depression Scale (HADS) [36], a widely used and validated instrument for detecting psychological distress in ICU survivors [37,38]. The HADS comprises two subscales, one for anxiety and one for depression, each consisting of seven items rated on a 4-point Likert scale, with a maximum score of 21 points per subscale [36]. The instrument has been widely used and shown to be a valid measure of psychological distress in ICU survivors [37]. The instrument has demonstrated good internal consistency and construct validity in the ICU survivor population. A subscale score of $\geq$11 points was used to indicate clinically significant symptoms [36,39].
- Assessment of post-traumatic stress symptoms using the using the Post-Traumatic Stress Symptoms 14-Question Inventory (PTSS-14), a validated instrument commonly used to assess post-traumatic stress symptoms in ICU survivors [40,41]. Part A consists of four questions on traumatic memories answered yes or no, and Part B consists of 14 questions with structured statements rated on a 7-point Likert scale, rendering a maximum total score of 98 points. The PTSS-14 has demonstrated good internal consistency, test–retest reliability, and construct validity in critically ill populations [40,41]. A Part B score $>$ 45 points indicates substantial symptoms of post-traumatic stress [40].
- Assessment of pain using the Numeric Rating Scale (NRS), a widely used and validated instrument for assessing pain intensity in clinical settings [42–44]. Participants rated current pain at rest and during movement on a scale from 0 (no pain) to 10 (worst imaginable pain) [42,44]. The NRS has demonstrated good validity and reliability for assessing self-reported pain intensity and is recommended for use in critically ill patients who can communicate [43,45].

The selected outcome measures were chosen based on their widespread use in ICU survivorship research and their demonstrated validity and reliability in assessing physical, psychological, and cognitive outcomes in ICU populations. Based on the initial assessments, an individual multiprofessional rehabilitation plan was initiated immediately following the first visit and developed in collaboration with the participant. The plan was informed by the identified needs and aimed to guide early recovery after ICU discharge. Goals were documented and communicated to the ward staff to ensure continuity of care. The case manager conducted follow-up visits to support the realisation of the multiprofessional rehabilitation plan and facilitated consultations or referrals to relevant healthcare professionals as needed. Before hospital discharge, the case manager confirmed that a structured plan for continued rehabilitation was established.

Out-of-hospital follow-up.

After discharge, the case manager conducted weekly telephone follow-ups for 12 weeks using a structured questionnaire (Table 2.) to assess psychological and physical well-being, adherence to the rehabilitation, and the need for coordination with primary care.

Outcome measures

Process feasibility of the study methods

The process feasibility of study methods was assessed through recruitment, retention and assessment completion. Recruitment was evaluated by calculating the number of enrolled patients relative to the total number of eligible participants. Retention was evaluated by determining the proportion of patients who completed the study. Assessment completion was defined as the total number and response rate (%) of completed assessments across all participants.

Table 2
Out-of-hospital telephone follow-up guide.

Out-of-hospital telephone follow-up	Attended (numbers)	Adherence rate (%)	Answers (numbers)
	37/40	93%	
Physical rehabilitation			
"Are you currently undergoing rehabilitation?"	37	100%	Yes $n = 17$, No $n = 20$
If No: "Do you feel a need for rehabilitation?"	20	100%	Yes $n = 5$, No $n = 15$
If Yes: "Are you satisfied with your rehabilitation, and are you receiving the support you need?"	16	94%	Yes $n = 16$, No $n = 0$
"Have you regained your pre-illness level of physical functioning?"	37	100%	Yes $n = 12$, No $n = 25$
If No: "Do you feel that you can reach that level of physical ability on your own, without help?"	25	100%	Yes $n = 10$, No n = 15
Psychological wellbeing			
"Do you feel a need for emotional support through conversation?"	37	100%	Yes $n = 3$, No $n = 34$
Cognitive function			
"Are you experiencing difficulties with memory, concentration or mental fatigue?"	36	97%	Yes n = 10, No $n = 26$
If Yes: "Are you in contact with healthcare for these issues, and do you feel you are getting the help you need?"	10	100%	Yes n = 3, No $n = 7$
"Are you able to perform your daily activities (e.g., dressing, bathing, cooking) in the same way as before your illness?"	37	100%	Yes n = 25, No n = 12
Pain (NRS)			
"Please indicate your current level of pain at rest on a scale from 0 to 10. 0 means no pain and 10 means the worst imaginable pain."	26	70%	Yes n = 3, No $n = 23$
"Please indicate your current level of pain during movement on a scale from 0 to 10. A score of 0 indicates no pain, while 10 represents the worst imaginable pain."	26	70%	Yes n = 4, No $n = 22$

Abbreviation: NRS = numeric rating scale.

Feasibility of the intervention programme

The intervention's feasibility was evaluated through the acceptability of the follow-up programme from the perspective of the participants/families, healthcare professionals, and safety monitoring.

The experiences of participants and families were obtained from telephone interviews after the intervention and three months after ICU discharge. The interviews aimed to elicit reflections upon the intervention's relevance, perceived benefits, challenges and suggestions for improvement. The interview guide is provided in Supplementary File 2. The interviews were audio-recorded and transcribed verbatim using automated transcription software. Interviews were conducted in Swedish and subsequently translated into English for analysis; all translations were reviewed by a professional translation service to ensure linguistic accuracy.

To evaluate the ward staff's experiences, a study-specific questionnaire was distributed to those involved in the care of the included patients. This questionnaire consisted of nine study-specific items related to collaboration, communication, knowledge transfer and perceived benefits for participants and their families. Provided in Supplementary File 3.

Safety monitoring

Assessment of potential harm from the intervention was conducted. Monitoring of vital signs both during and after the physical interventions was a standard procedure. Any adverse events were documented in the participant's medical chart and reported.

Fidelity of the intervention

Intervention fidelity refers to how well participants adhere to the intended components of an intervention [46]. In this study, assessed by recording the total number of completed sessions relative to the number of planned sessions. Fidelity was also addressed by continuously collecting the case managers' experiences of intervention delivery throughout the study period. Feedback was documented in informal fieldnotes and structured reflections by the research team during regular debriefing sessions. At the closure of the study, case managers were invited to participate in a reflective feedback process, where they were encouraged to elaborate upon their experiences, suggest improvements and comment on the feasibility and acceptability of the intervention components.

Patient-reported outcome measures three months after ICU discharge

Three months after ICU discharge, follow-up questionnaires were distributed to patients via email or postal mail. One reminder was sent to non-responders. The 3-month time-point was selected as it is commonly used in studies of PICS to capture early, yet clinically relevant physical and psychological morbidity after ICU discharge [47]. The timing is also consistent with current guideline recommendations for post-ICU follow-up [48].

The questionnaires included HADS, PTSS-14, Barthel Index, NRS and self-reported physical activity using the NBWH-PA. Participants also received the RAND-36, covering eight domains of health-related quality of life (HRQoL): physical functioning (10 items), role physical (4 items), bodily pain (2 items), general health (5 items), vitality (4 items), social functioning (2 items), role emotional (3 items) and emotional well-being (5 items) [49]. The scores range from 0 to 100, with higher scores indicating a better HRQoL [50]. The RAND-36 has demonstrated good validity and reliability in both general and Swedish populations and has been applied in studies of survivors of critical illness [50,51].

Sample size

We aimed for a sample size of approximately 60 patients, considered sufficient to assess intervention feasibility and fidelity [52]. The study included quantitative and qualitative outcomes, process-related measures such as recruitment and retention rates. Given that the different components of the study require different sample sizes, we prioritised pragmatism to ensure that all relevant aspects could be assessed. Rather than relying on the estimation of a predetermined sample size, we focused on transparently reporting the rationale behind our design choices, to support meaningful interpretation.

Data analysis

Quantitative data

Participants' demographics, clinical characteristics, feasibility outcomes and patient-reported outcome measures were reported descriptively, with medians and interquartile ranges (IQR) for continuous variables and counts and percentages for categorical variables. All statistical calculations were performed using IBM SPSS version 30 for Windows.

Qualitative data

The follow-up interviews were analysed using an inductive method inspired by Elo and Kyngäs [53]. Data from the semi-structured interviews were systematically analysed and organised into descriptive categories to address the study's feasibility. The interviews were conducted by MJ, a trained member of the research team with experience in

qualitative methods and had a mean duration of 20 min. Two researchers (KS, MJ) independently read and re-read the transcripts to gain a comprehensive understanding of the data and identify units of meaning relevant to the research aim. Open coding was performed together with a third researcher (AS), and codes were grouped into categories based on similarities and patterns. A coding sheet was developed to support consistency and transparency. The data were categorised based on the acceptability of the intervention, including factors that facilitated or hindered.

Results

Process feasibility of the study methods

A total of 382 patients were eligible for risk screening; however, screening was not conducted for 227 patients. Among the 155 patients screened for risk of physical and/or psychological problems at ICU discharge, 57 (37%) were identified as being at increased risk, and 41 (72%) of these eligible patients consented to participate. See Fig. 1 for recruitment details and Table 3 for participant characteristics.

Of the 41 participants, 37 completed the follow-up programme, resulting in a retention rate of 90% and a dropout rate of 10%. Of these 37 participants, 22 (59%) returned the follow-up questionnaires (Table 4 and Fig. 1.).

Feasibility of the intervention programme

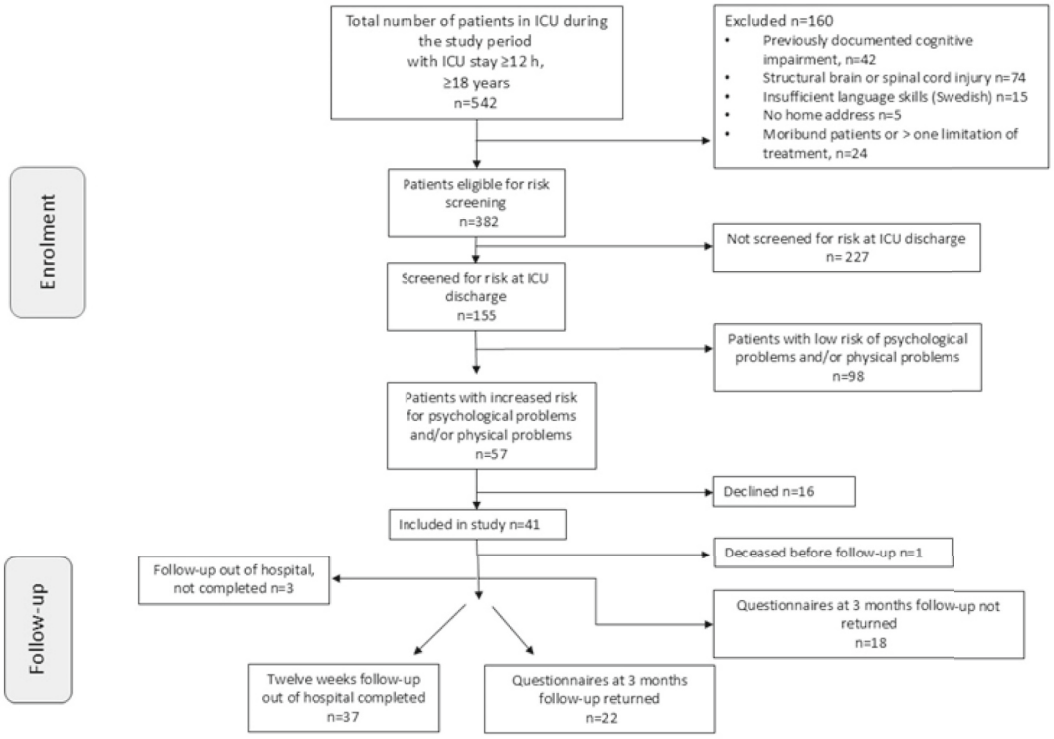
The acceptability of the intervention for patients and their families was assessed through follow-up interviews. Findings are organised into three main categories: Perceived benefits, Facilitators, and Barriers with ten subcategories. For an overview, see Fig. 2.

Table 3
Patient characteristics.

Variable	Included patients (n = 41)
Age in years, median (IQR)	66 (59–74)
Sex, n (%)	
Male	22 (54)
Female	19 (46)
Level of education, n	
Elementary school	8
Senior high school	12
University	21
Family situation, n	
Married/cohabiting	20
Single	11
Children at home	4
Missing information	10
Previous psychological problems, n	
Yes	15
Missing	1
Pre-ICU Barthel Index, median (IQR)	100 (95–100)
Admission diagnosis*	
Infection	15
Surgical complications	10
Cardiovascular	10
Trauma	5
Obstetrics	2
Other	5
SAPS, median (IQR)	56 (48–66)
Mechanical ventilation, n (%)	22 (54)
ICU length of stay (hours), median (IQR)	87 (47–198)

* More than one answer could be selected.

Patients reported that ward-based follow-up visits enhanced their sense of safety and continuity, promoting engagement in rehabilitation.



Abbreviations: ICU= intensive care unit

Fig. 1. Flow chart of enrolment and follow-up Abbreviations: ICU = intensive care unit.

Table 4
Patient-reported outcome measures three months after ICU discharge.

Variables	Respondents Numbers (% of included participants)	Results (Medians (IQR)
HRQoL, RAND-36	21 (55)	
Physical functioning		63 (30–85)
Physical role functioning		45 (45–100)
Bodily pain		25 (0–75)
General health		50 (35–70)
Vitality		50 (40–65)
Social functioning		63 (50–100)
Emotional role functioning		67 (0–100)
Mental health		76 (56–84)
Psychological symptoms		
HADS Anxiety	22 (55)	5 (3–8)
HADS-A ≥ 8	6 (27)	
HADS Depression	22 (55)	4 (2–7)
HADS-D ≥ 8	4 (18)	
PTSS-14	21 (53)	27 (23–39)
PTSS-14 ≥ 45	4 (19)	
ADL Barthel Index	20 (50)	100 (85–100)
Pain assessment, NRS		
Pain at rest	19 (46)	1 (0–5)
Pain during activity	9 (22)	1 (0–3)

Abbreviations: ADL = Activities of Daily Living, HADS = The Hospital Anxiety and Depression Scale, HRQoL = Health-Related Quality of Life, NRS=Numeric rating scale, PTSS=Post-Traumatic Stress Symptoms.

The multiprofessional approach offered a holistic understanding of needs and supported motivation. Meeting different professionals helped patients clarify treatment, reflect on their experiences, and build trust. Psychosocial support aided emotional recovery and understanding of their trajectory. Overall, the intervention was considered acceptable and meaningful, supporting both physical and psychological recovery.

Several implementation facilitators and barriers were identified by both the patients and their families. Key facilitators were the multiprofessional approach, continuity through familiar staff, and the involvement of family members providing emotional and practical support. Compassionate ward staff further enhanced patients' sense of safety and engagement. Reported barriers included fatigue and low energy—often linked to inadequate nutritional intake due to poor appetite or nausea—as well as insufficient pain relief despite medication. Family members also experienced limited information about the intervention and inadequate emotional support, which hindered their ability to assist.

Acceptability from the perspective of healthcare professionals

Twenty-one ward-based healthcare staff completed the questionnaire. Overall, they found the intervention acceptable, and the time required appropriate. Thirteen staff members (62%) reported improved interdisciplinary collaboration and communication and noted better understanding of patients' post-ICU needs. Most respondents, 86%, felt the follow-up programme benefited patients. Detailed information in Supplementary File 3.

Safety

No adverse events were reported by patients, family members or case managers.

Fidelity of the intervention

All 41 participants received a visit from the case manager within 1–5 days following ICU discharge. On average, the case manager spent approximately 30 min per initial visit. A mean of three follow-up visits per participant was delivered by the case manager. The total number of completed sessions relative to the number of planned sessions for each intervention component is reported in Table 1.

During the initial ward visit, most participants underwent delirium, psychological, pain and functional assessment, and almost all received tailored physiotherapy. Based on identified needs, participants were referred to and received input from multiple healthcare professionals, including physiotherapists, occupational therapists, social workers, and speech therapists, reflecting the multiprofessional nature of the intervention (See Table 1). Physiotherapeutic rehabilitation was subsequently initiated and continued throughout the hospital stay.

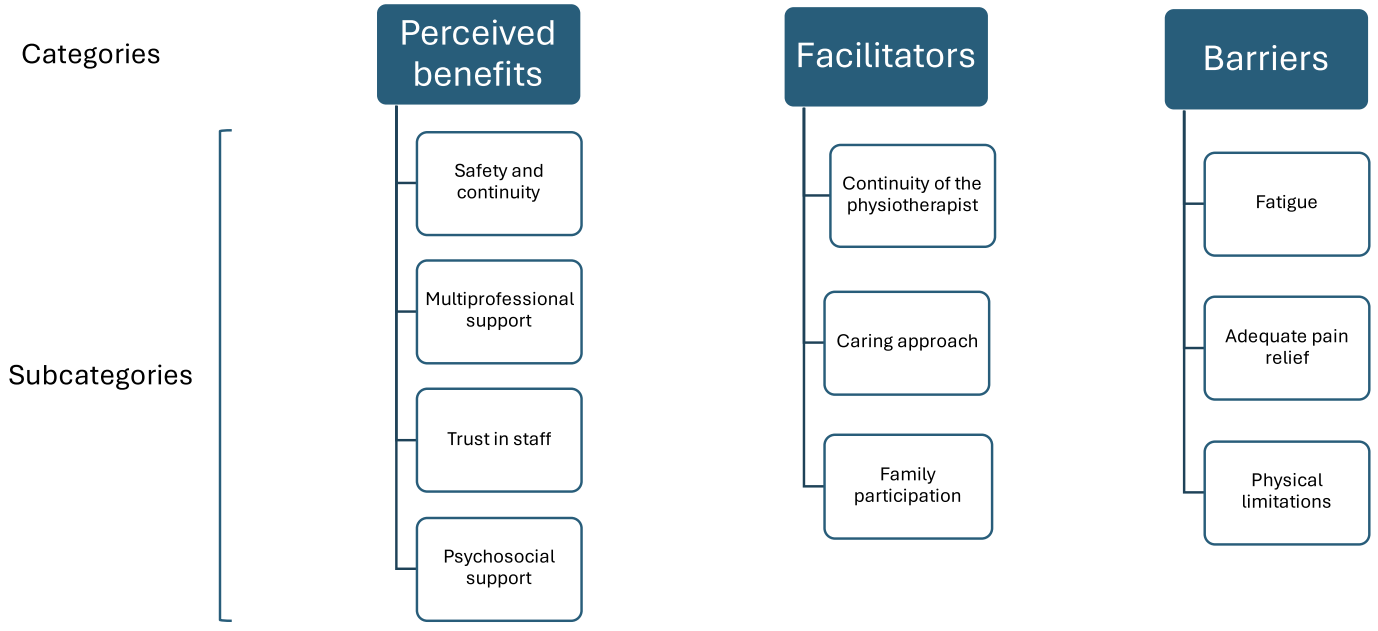


Fig. 2. Categories and subcategories.

Out-of-hospital follow-up

One participant died before hospital discharge, and three were lost to follow-up. Among the remaining 37 (93%), a total of 161 telephone calls were made during the 12-week follow-up period. At the end of each call, a timepoint for the next contact was agreed upon. Frequency adjustments were made based on individual preferences, rather than adhering rigidly to the originally projected weekly schedule. At the final follow-up call, 17 participants (43%) reported ongoing physical rehabilitation. Twenty-five participants (63%) had not regained their pre-illness physical function. For detailed information, see Table 2.

The case managers' experiences of intervention delivery

According to the case managers, the follow-up programme was generally acceptable, despite several challenges. Delivery was occasionally hindered by participants' clinical deterioration, scheduling issues, competing care needs, or patients declining to participate. The case managers observed that participants often underestimated their functional abilities, highlighting a gap in self-awareness early in recovery. Collaboration with ward teams was considered vital for tailoring rehabilitation and ensuring continuity. Pain assessment focused mainly on its impact on mobilisation, rather than formal evaluation, which the case managers later regarded as a limitation.

Patient-reported outcomes three months after ICU discharge

Three months after ICU discharge, follow-up questionnaires were sent to participants, of whom 22 (55%) returned the questionnaires. Response rates and results are presented in Table 4. Twenty-one participants completed the RAND-36. The lowest scores were observed in Bodily Pain (median 25, IQR 0–75) and Physical Role Functioning (median 45, IQR 45–100), followed by General Health (median 50, IQR 35–70) and Vitality (median 50, IQR 40–65).

Four out of 22 (18%) responders reported clinically significant symptoms of anxiety and/or depression and 4/19 (21%) reported clinically significant symptoms of post-traumatic stress. The Barthel Index was completed by 20 participants (50%), with a median score of 100 (IQR 85–100). The median NRS score was 1 both at rest and during movement, based on 19 responses (IQR 0–5) and 9 responses (IQR 0–3), respectively. Twenty participants (50%) responded to the NBHW-PA questions. Of these, 13 reported engaging in 0–30 min/day of physical exercise (e.g. running, aerobics, or ball sports), and 11 reported less than 150 min/week of general physical activity (e.g. walking, cycling, or gardening). Ten out of 20 participants had a total score of <11, indicating that they did not meet the recommended level of physical activity [37].

Discussion

The intervention was assessed as feasible, with a 90% retention rate, and considered acceptable by participants, family members and healthcare professionals; however, incomplete implementation of the screening process in this pragmatic setting indicates a risk of selection bias and highlights the need for improved integration of screening procedures. Rehabilitation adherence was high; 33 out of 41 patients participated in developing an individualised plan. Case managers reported that physical interventions were individually tailored yet sufficiently challenging, supporting engagement and meaningful progress. A balance between individualisation and sufficient therapeutic challenge appears essential for sustaining adherence and achieving meaningful outcomes. The strong engagement in rehabilitation planning suggests that personalised approaches may enhance patients´ sense of ownership and motivation. Similar findings have been reported with patients highlighting encouragement and recognition of individual needs as key to recovery [54].

Adherence was high for the first ward visit, establishing a rehabilitation plan, and follow-up calls. All patients received their first visit within five days, but adherence to specific components ranged widely (15% for diary review to 93% for delirium assessment), reflecting the intervention's complexity and the need for personalised care. This has also been described in fidelity studies involving other patient populations with complex needs [55]. Follow-up calls (completed for 93% of participants) helped identify unmet needs, particularly related to physiotherapy and care coordination.

A full CPAx assessment during the first ward visit was often not feasible due to patients' fatigue. Physical assessment was successfully assessed during subsequent physiotherapy sessions, underscoring the need for flexible timing. While CPAx was not designed for ICU aftercare, its use aligned with a previous study, demonstrating its ability to track physical progress in ward settings, despite some ceiling effects [56]. Importantly, the first five CPAx items are part of the physical risk screening instrument used in this study [18], making it advantageous to use the same instrument throughout the rehabilitation pathway, supporting consistent data collection and smoother care transitions.

Nearly 39% of participants were at risk of both physical and psychological problems, highlighting the need for a multiprofessional model. The presence of pre-existing psychological issues in 37% further emphasises the importance of early psychosocial screening and intervention. These findings align with literature emphasising the need for integrated care pathways and comprehensive support for ICU survivors [57], with multiprofessional models essential for meeting the complex rehabilitation needs of patients with PICS [58].

Limitations

This study was designed as a pragmatic feasibility study, in which the responsibility for risk screening was integrated into routine clinical practice and performed by ICU staff. Although 382 patients met the eligibility criteria, screening was not conducted in 227 cases. This reflects challenges in the implementation of the screening procedure in a real-world clinical setting. In particular, the screening instrument was not fully embedded in clinical workflows, and its use depended on staff availability, prioritization, and familiarity with the study procedures. Consequently, non-screening did not occur systematically based on patient characteristics but was largely influenced by organisational and practical factors. We acknowledge that this introduces a risk of selection bias, as included patients may not be fully representative of the broader eligible population. This limitation affects the interpretation of feasibility, and our findings suggest that successful implementation requires more substantial integration of the screening process into routine care, including staff training, clearer workflows, and potentially automated or protocol-driven screening procedures.

The timing of the initial follow-up visit varied between 1 and 5 days after ICU discharge, which may have influenced the intervention exposure and subsequent outcomes. Although this variation reflects real-world clinical practice and logistical considerations in delivering early post-ICU care, differences in timing may have affected patients' readiness to engage in rehabilitation and the ability to identify early deficits. The timing may therefore be considered in the design of the subsequent randomised controlled trial.

Excluding patients with insufficient Swedish language skills may have reduced sample diversity. Conducting the study at two hospitals within one region also restricts external validity, although both the in-hospital and out-of-hospital components appear feasible and adaptable to other settings. The ≥12-h ICU length of stay threshold was chosen as a pragmatic compromise to exclude short, low acuity observational admissions while retaining clinically meaningful ICU exposure. However, this cutoff differs from the ≥24-h (or longer) threshold used in some prior studies [59], which may affect comparability and cohort composition.

Although CPAx is validated for critically ill patients, its suitability for

general ward populations needs further evaluation. The distribution of follow-up questionnaires by email or post may have influenced response rates; while email is more cost-effective, postal surveys generate higher returns [60]. Future studies may benefit from offering both options, optimising timing and reminders.

Conclusion

This study demonstrated that implementing a multiprofessional follow-up programme for ICU survivors at risk of physical and psychological problems is feasible. Feedback from patients, family members, ward staff, and case managers informed several proposed adjustments to improve the operational delivery of the intervention while preserving its core principles. Adjustments were primarily logistical and did not affect the fundamental assumptions. Based on the results, a definitive trial to determine the effectiveness of the programme among high-risk ICU patients would be feasible with adaptations to improve the screening and assessment completion.

CRediT authorship contribution statement

Katinka Siesage: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Anna Schandl:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Eva Joelsson-Alm:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Matheo Johansson:** Writing – review & editing, Investigation, Formal analysis, Data curation. **Emelie Karlsson:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation. **Anna Milton:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Clinical trial registration

Clinicaltrials.gov, NCT 06118606.

Ethics statement

The study was approved by the Swedish Ethical Review Authority (No: 2022–03436-01 with the amendment: 2023–08073-02) and adhered to the Declaration of Helsinki and its later amendments (World Medical Association Declaration of Helsinki: ethical principles for medical research involving human subjects, 2024). All participants gave written informed consent. The study was registered at clinicaltrials.gov, NCT 06118606.

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Declaration of competing interest

The authors declare that they have no competing interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.iccn.2026.104492.

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